Multilingual Voice Interaction System
for Humanoid Robot (Unitree G1)

- Real-time embedded conversational pipeline
- Jetson Orin constraints
- Designed & deployed on real hardware

Problem Statement

- Enable natural voice interaction
- Vietnamese + Japanese (target)
- Multi-speaker environment
- Edge deployment (Jetson Orin)
- No long-term API dependency

System Constraints

- Audio I/O via Unitree SDK
- No direct ALSA control
- Limited CUDA multithreading
- Cloud latency impact
- Target response < 2 seconds

System Architecture

- Mic → Wakeword → VAD + ASR
- Intent Cache → LLM + RAG
- TTS → Audio Normalization → Speaker
- Client (Jetson) + Server (Cloud/Dev)

Audio Engineering Challenges

- Sample rate mismatch
- Buffer timing misalignment
- Packet loss & distortion
- Amplitude threshold tuning
- Wave normalization solution

Wakeword & ASR Strategy

- Vosk wakeword (Vietnamese support)
- Future: Porcupine custom wakeword
- VAD + silence detection
- Sentence boundary timing control

LLM & RAG Pipeline

- GPT-4o for reasoning
- e5 multilingual embedding
- FAISS index (~3–5GB)
- Intent cache optimization
- Latency: 1–2s cached | 5–15s dynamic

Embedded Optimization

- Edge vs Cloud workload split
- Auto network detection
- Bash auto-start system
- Remote UI via ngrok
- Robust deployment strategy

Failure Cases & Limitations

- Multi-speaker failure
- High noise instability
- Japanese not fully deployed
- Cloud API dependency
- Latency still high

Technical Roadmap

- Local Whisper (quantized)
- Speaker diarization model
- Beamforming mic array
- On-device LLM (7B quantized)
- Reduce latency < 2s

Engineering Value

- Real-time robotic system design
- Embedded AI deployment
- Audio engineering depth
- Client–server robotics architecture